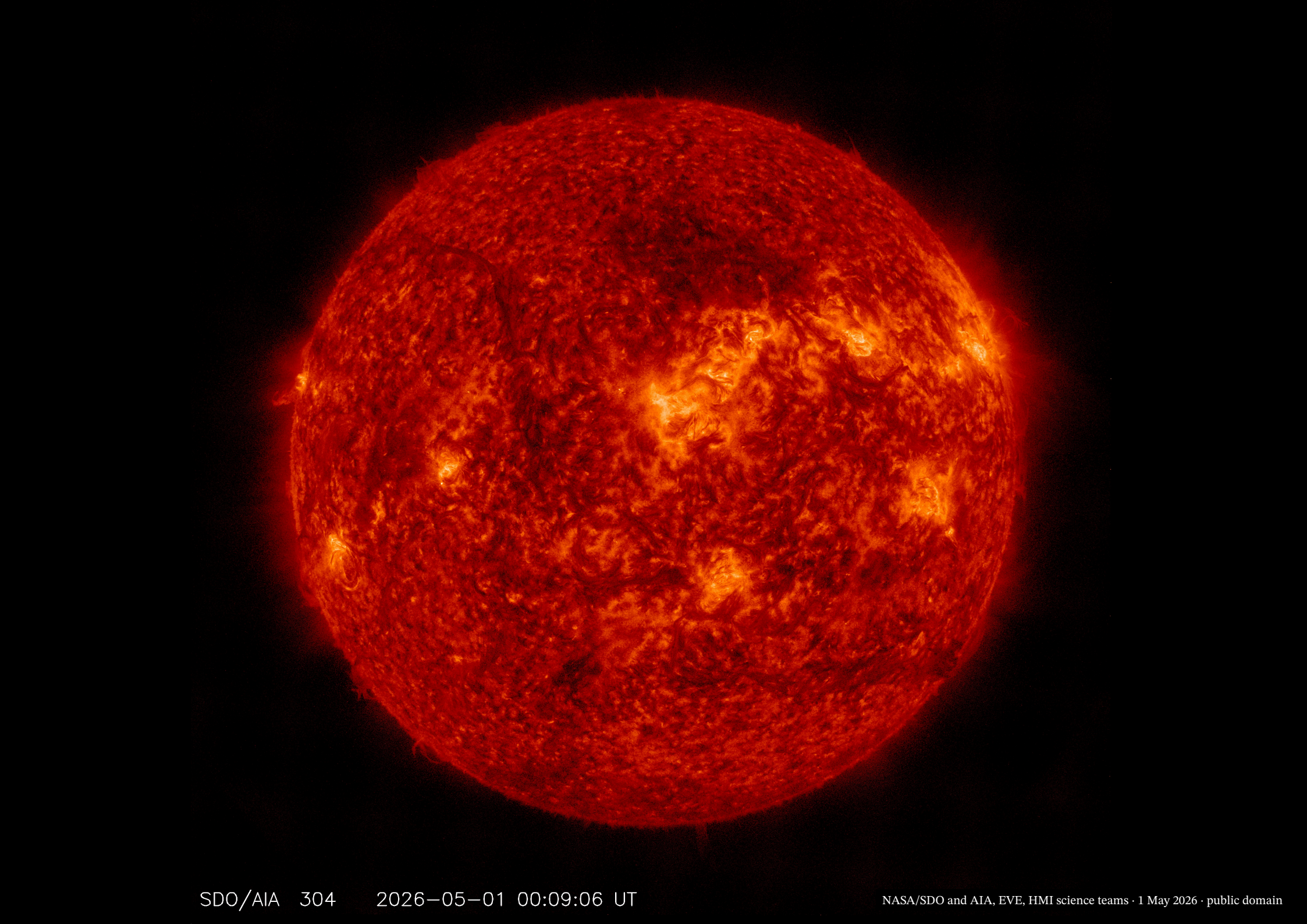
# Discovering fusion

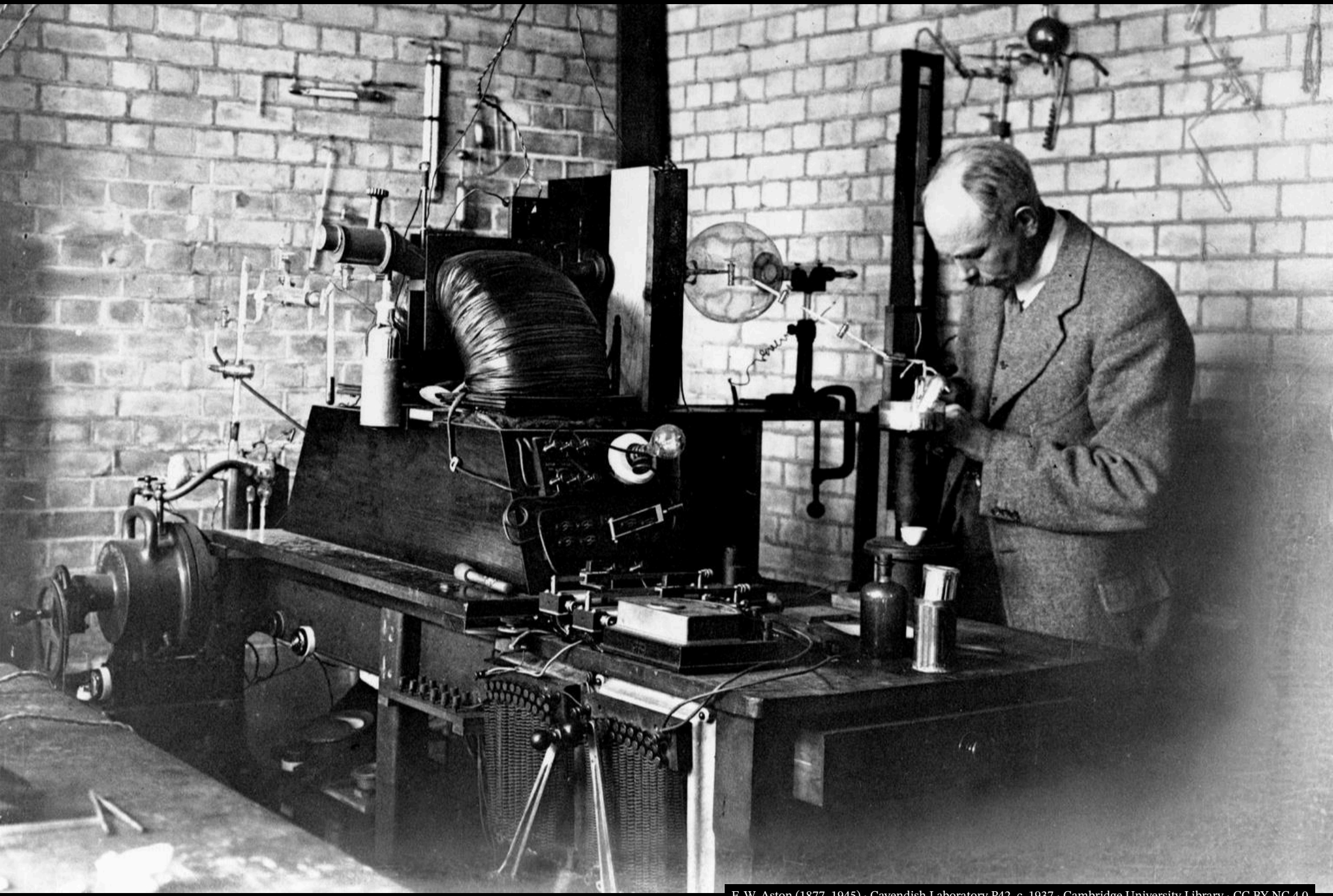
Christopher Albert




NASA/SDO and AIA, EVE, HMI science teams · 1 May 2026 · public domain

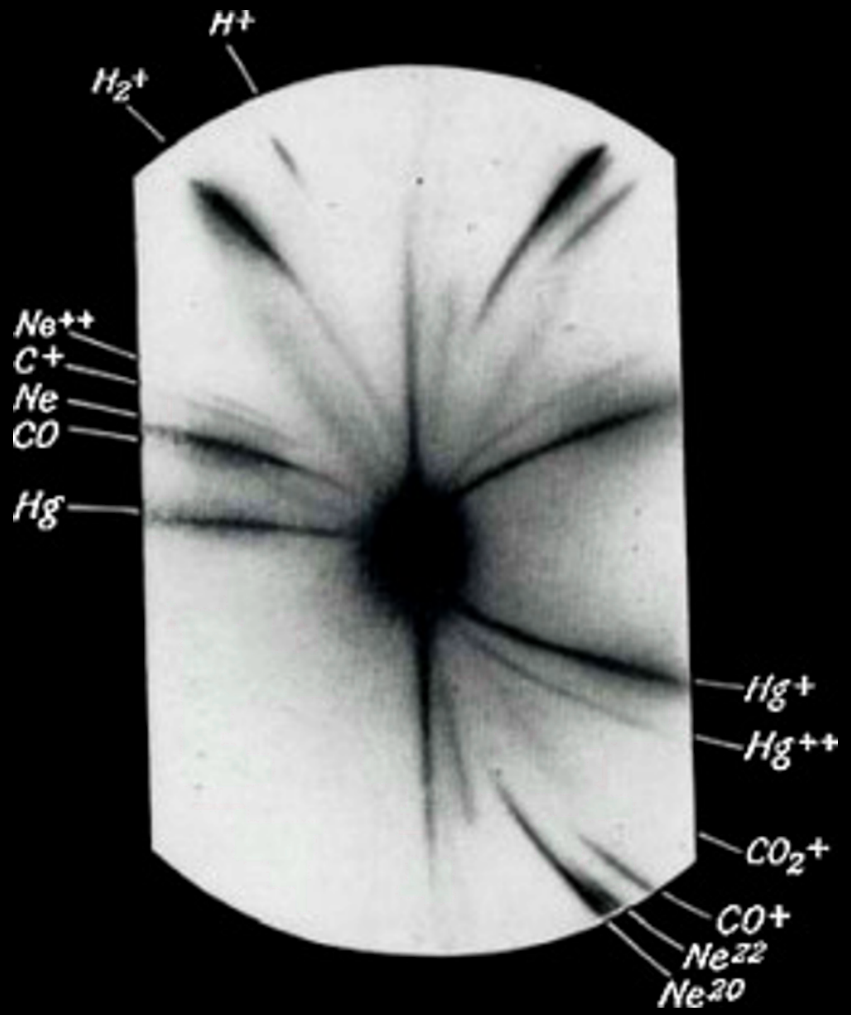


Wellcome Collection · M0003525 · wellcomecollection.org/works/h3cuknsg



F. W. Aston (1877–1945) · Cavendish Laboratory P42, c. 1937 · Cambridge University Library · CC BY-NC 4.0

# 1913




F. W. Aston · Isotopes (1922) · Historical discussion (2013)

# 1922




F. W. Aston · Isotopes (1922), Plate III

# Eddington · 1920

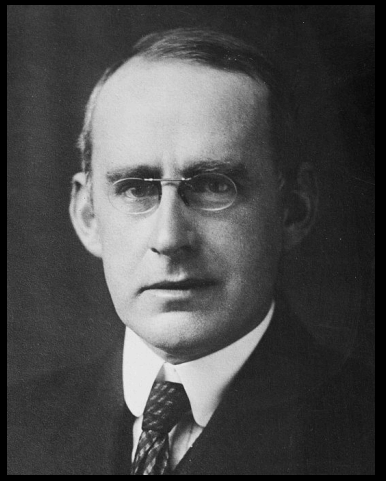
THE OBSERVATORY,

A MONTHLY REVIEW OF ASTRONOMY.

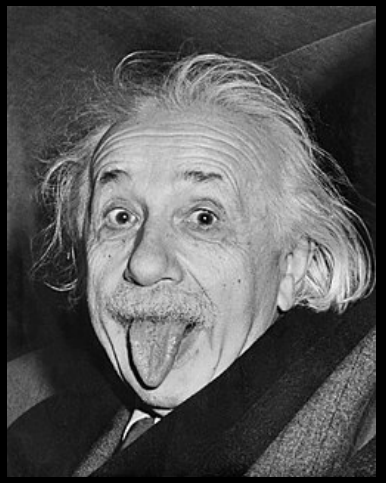
VOL. XLIII. OCTOBER, 1920. No. 557.

***Aston** has further shown conclusively that the mass of the helium atom is less than the sum of the masses of the four hydrogen atoms which enter into it and in this at least the chemists agree with him. There is a loss of mass in the synthesis amounting to 1 part in 120, the atomic weight of **hydrogen being 1.008 and that of helium just 4.00**... Now mass cannot be annihilated and the deficit can only represent the **mass of the electrical energy** liberated when helium is made out of hydrogen. **If 5% of a star's mass consists initially of hydrogen atoms**, which are gradually being combined to form more complex elements, the total heat liberated will more than suffice for our demands, and we need look no further for the source of a star's energy.*

*If, indeed, the subatomic energy in the stars is being freely used to maintain their furnaces, it seems to bring a little nearer to fulfillment our dream of controlling this latent **power for the well-being of the human race**—or for its suicide.*

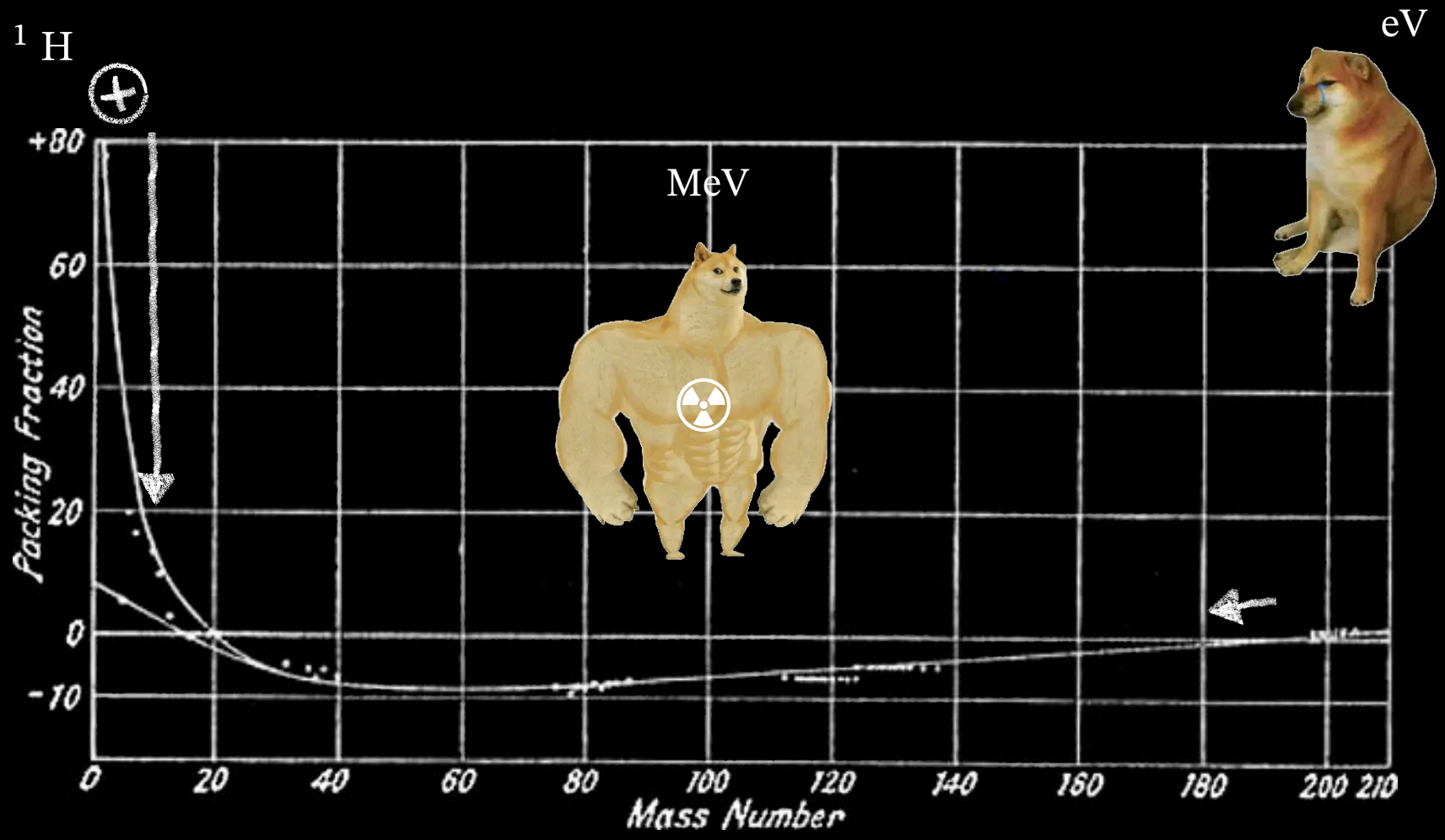


Arthur Eddington
(1882–1944)



Eddington · The Observatory 43, 341 (1920) · Einstein photograph: Arthur Sasse (1951) · portraits: FusionIntroSlides

# 1927




Source: Christopher Albert · FusionIntroSlides, slide 11 · original annotations and composition

# Atkinson and Houtermans · 1929

## On the possibility of building up the elements in stars

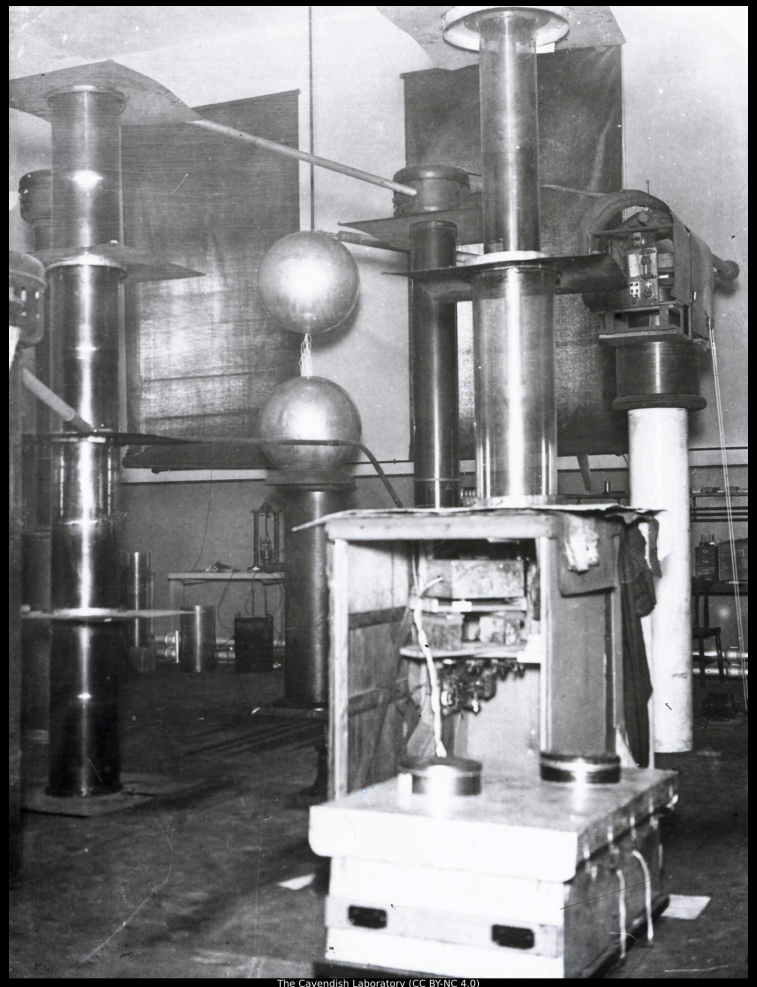
R. d’E. Atkinson · F. G. Houtermans · 1929

*The **quantum-mechanical** probability that a **proton penetrates an atomic nucleus** is calculated using **Gamow’s** method. It is found that, under the temperature and density conditions inside stars, **protons, but not α particles, penetrate** lighter elements sufficiently often to make the **building up of these elements** there appear probable. This makes it possible to explain the **energy production of stars through the mass defects** of the elements, while avoiding the assumption of six-particle collisions for helium formation. Some further hypothetical considerations on the building up of heavier elements follow.*




Atkinson & Houtermans · Z. Phys. 54, 656–665 (1929) · Title and full abstract translated from the original · Koralm image: FusionIntroSlides

# 1934




*Transmutation Effects Observed with Heavy Hydrogen*

By M. L. E. Oliphant, Ph.D. (Messel Research Fellow of the Royal Society), P. Harteck, Ph.D., and Lord Rutherford, O.M., F.R.S.

(Received April 14, 1934.)

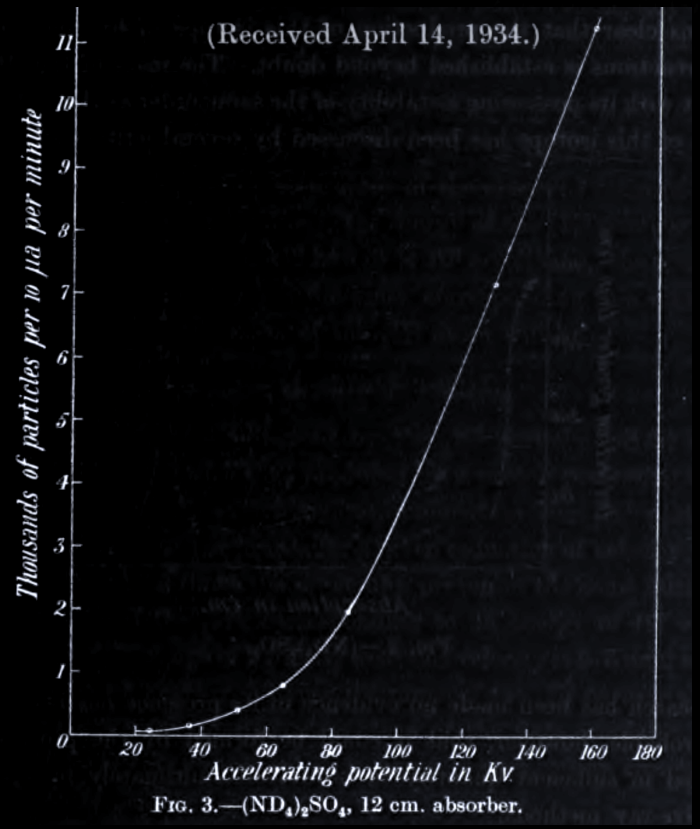
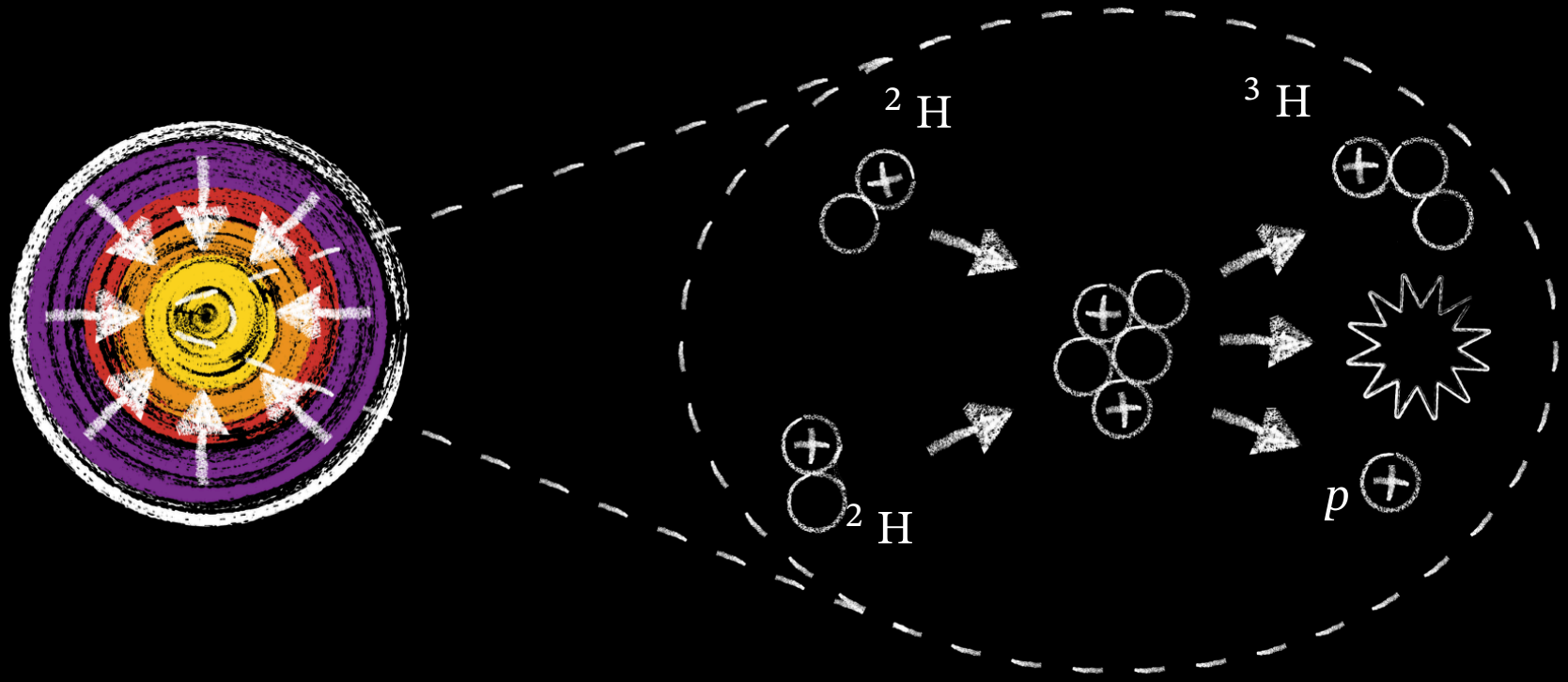



Fig. 3.—$(ND_4)_2SO_4$, 12 cm. absorber.

Cavendish P514 · Cambridge University Library · CC BY-NC 4.0 · Oliphant, Harteck & Rutherford (1934)

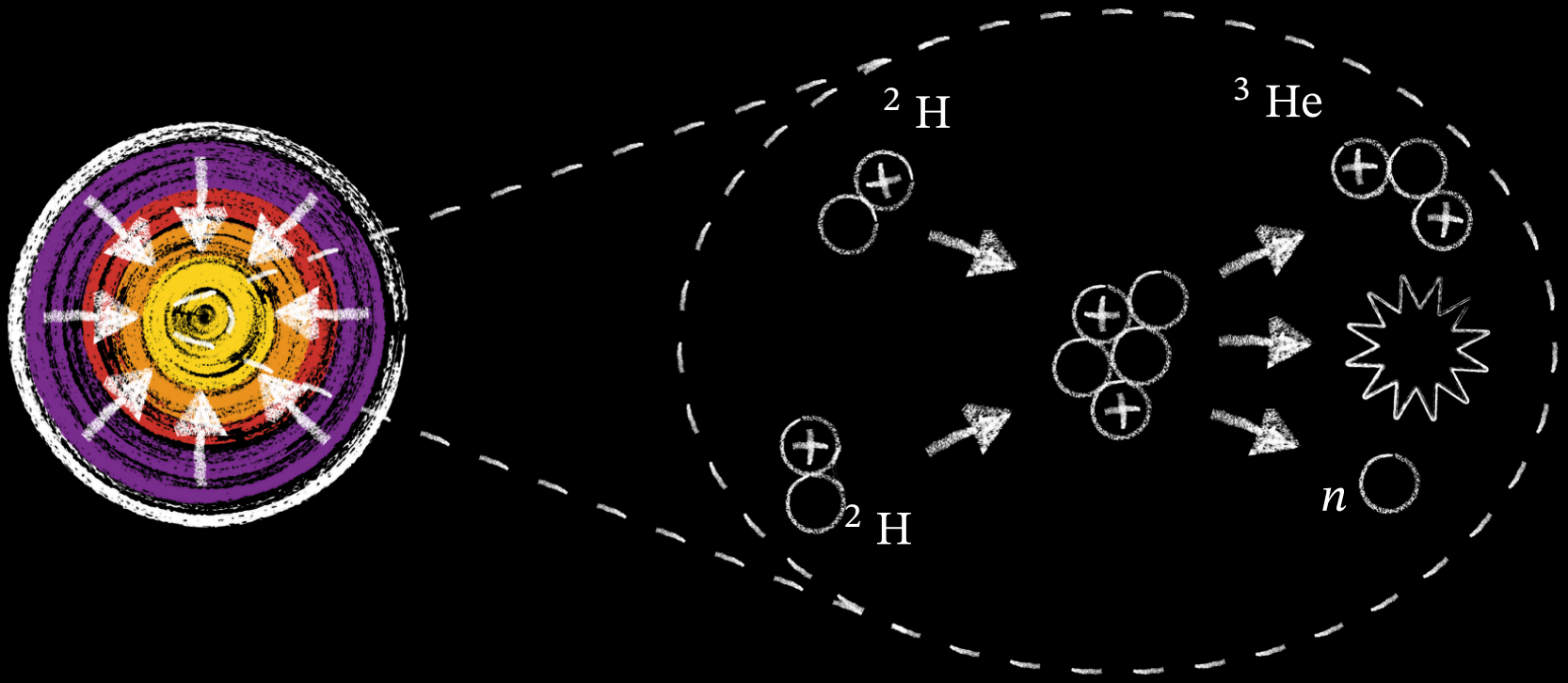
# $D + D \rightarrow T + p + 4.0\ MeV$




Christopher Albert · original reaction drawing, FusionIntroSlides · D: deuterium; T: tritium; p: proton; n: neutron

# $D + D \rightarrow T + p + 4.0\ MeV$

$D + D \rightarrow {}^{3}He + n + 3.3\ MeV$

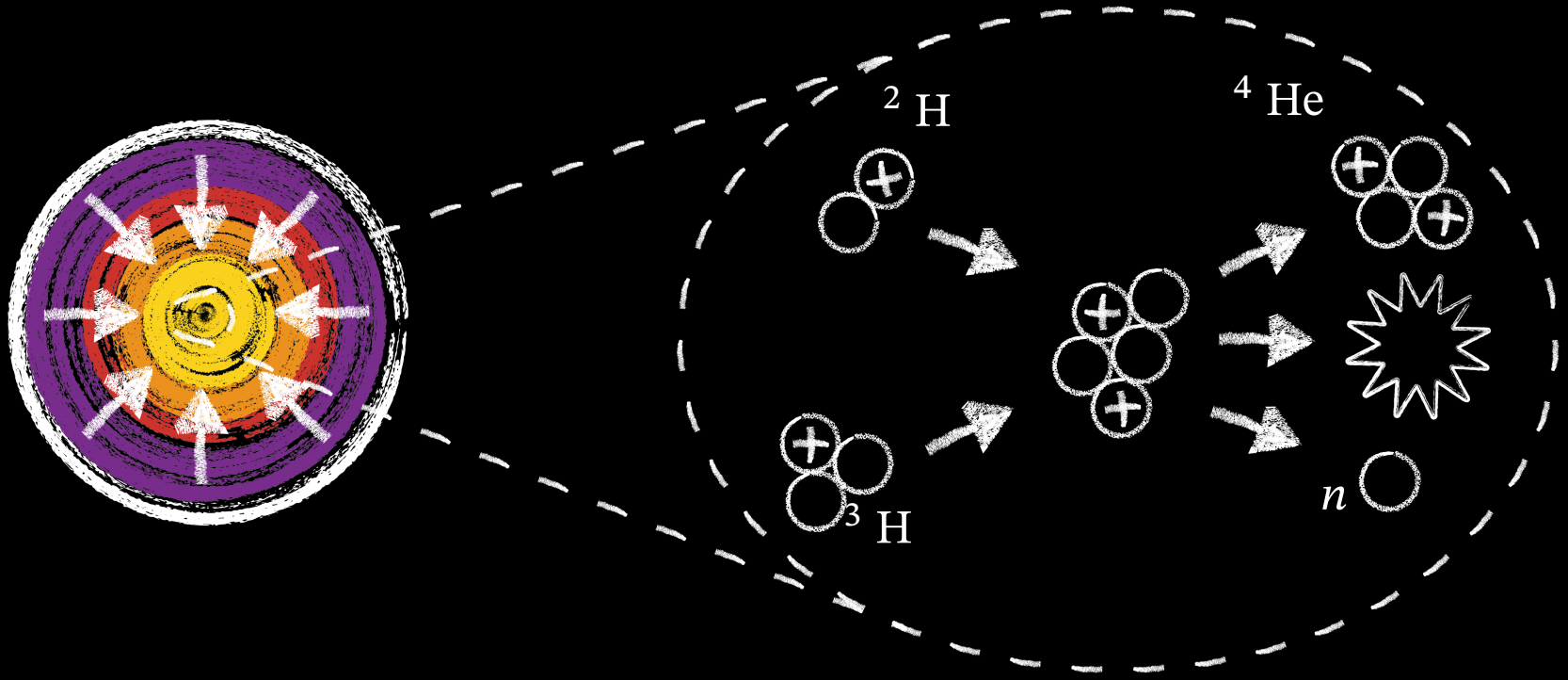



Christopher Albert · original reaction drawing, FusionIntroSlides · D: deuterium; T: tritium; p: proton; n: neutron

# $D + D \rightarrow T + p + 4.0\ MeV$

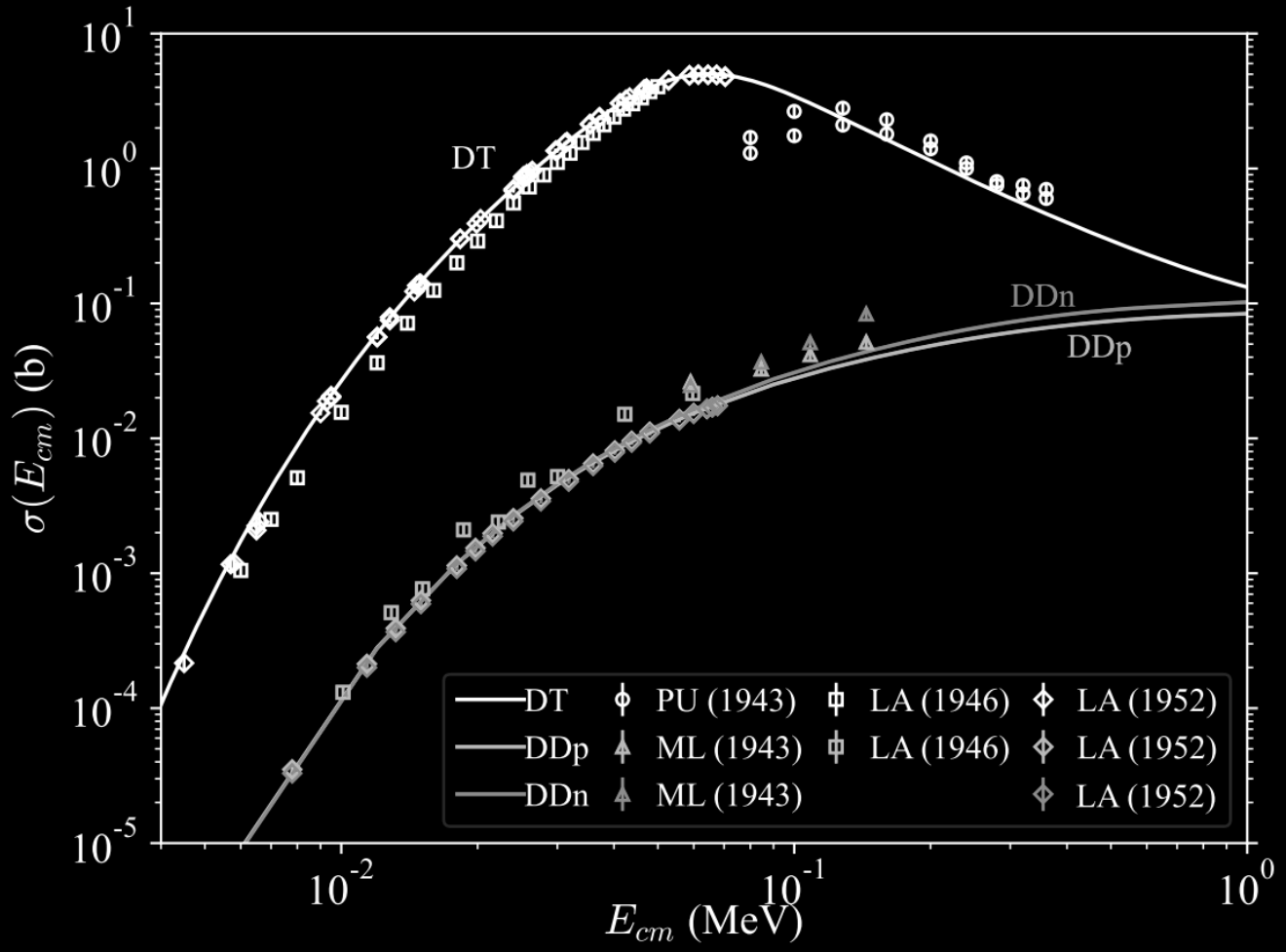
$D + D \rightarrow {}^{3}He + n + 3.3\ MeV$

$D + T \rightarrow {}^{4}He + n + 17.6\ MeV$




Christopher Albert · original reaction drawing, FusionIntroSlides · D: deuterium; T: tritium; p: proton; n: neutron

# 1943–1952




M. B. Chadwick et al. (2023), Fig. 2 · Historical fusion cross-section measurements

# Lawson · 1955



John David Lawson
(1923–2008)

$$n \times k_B T \times \tau_E \quad (p,\ \langle E \rangle)$$

$$p\tau_E > 3 \times 10^5 \text{ Pa s (D–T)}$$

Original diagram and portrait: FusionIntroSlides · Lawson (1957) · approximate D–T scale near optimal temperature

# The Sun

$\rho \sim 10^5 \mathrm{kg/m^3}$

$T \sim 10^7$ K

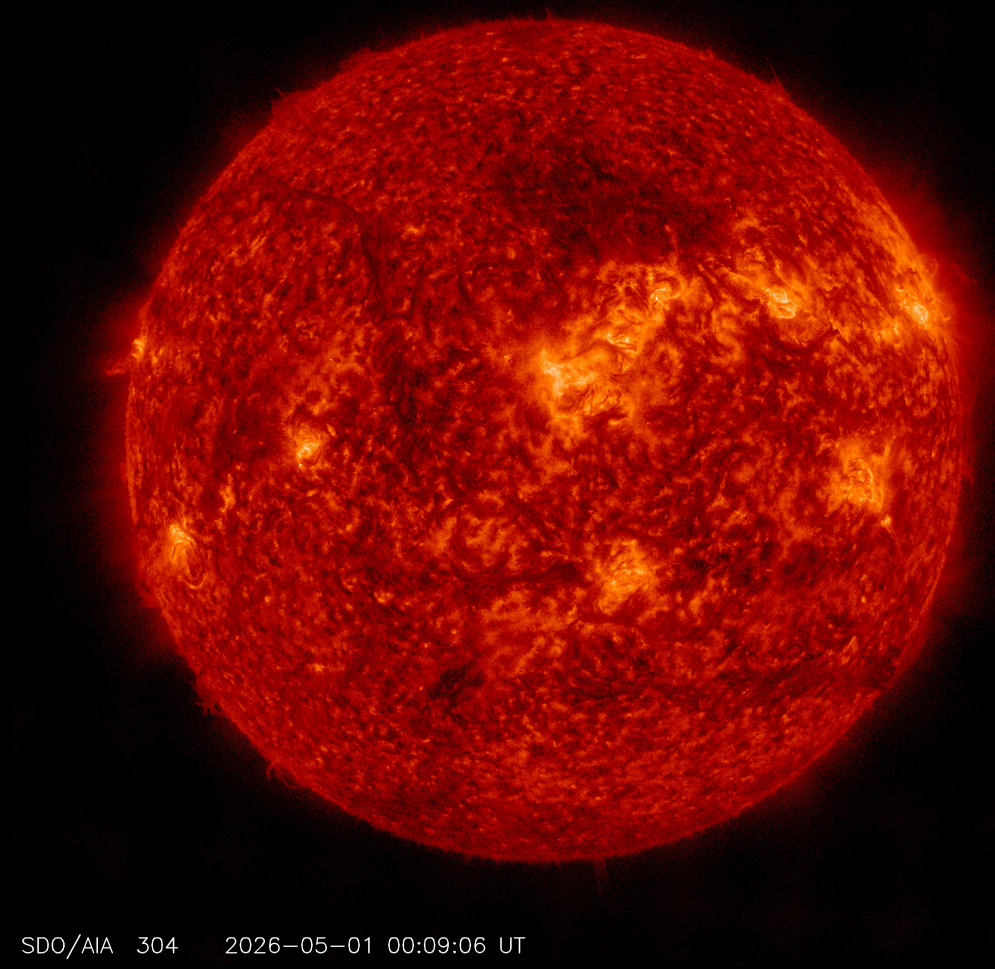
$p \sim 10^{16}$ Pa

$\tau_E \sim 10^7$ years $\sim 3 \times 10^{14}$ s

$p\tau_E \sim 3 \times 10^{30}$ Pa s

$> 10^{29}$ Pa s (p–p)

$4\,\mathrm{H} \rightarrow \mathrm{He} + Q$




FusionIntroSlides · Thermal time: solar stored energy / luminosity · NASA/SDO and AIA, EVE, HMI science teams · public domain

# Tokamak JET

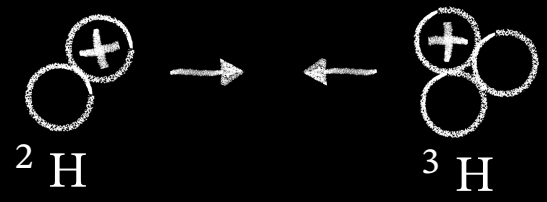
$\rho \sim 10^{-7} \mathrm{kg/m^3}$

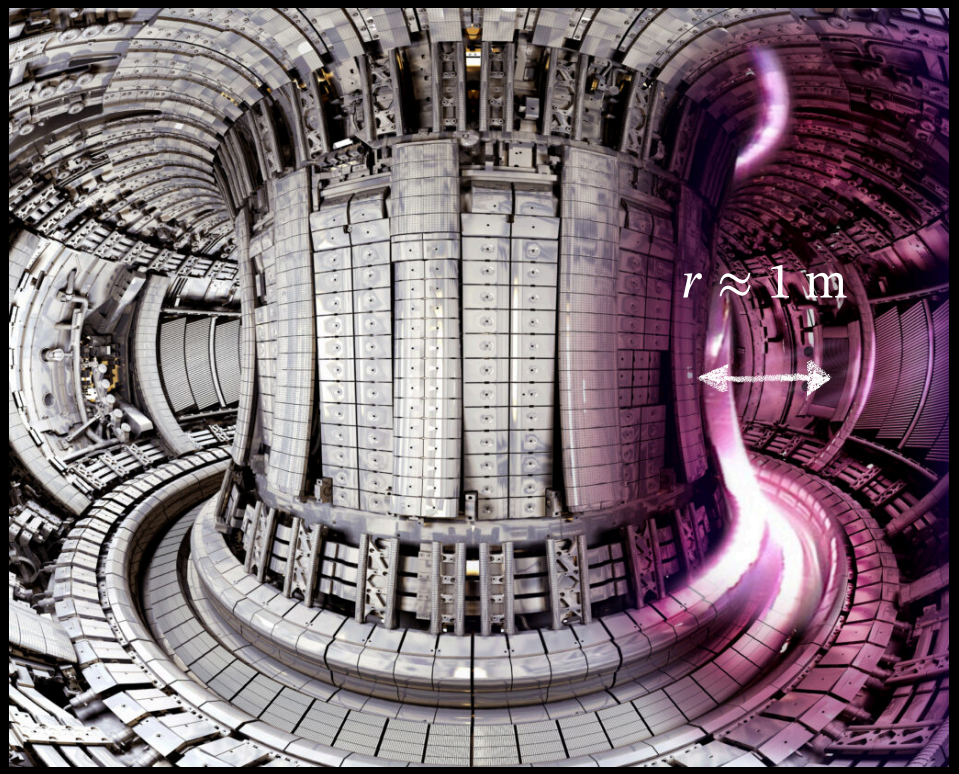
$T \sim 2 \times 10^8$ K

$p \sim 10^5$ Pa

$\tau_E \sim 1$ s

$$p\tau_E \sim 10^5 \text{ Pa s}$$




JET photograph: FusionIntroSlides, slide 21 · original annotations: Christopher Albert · approximate illustrative parameters

# National Ignition Facility

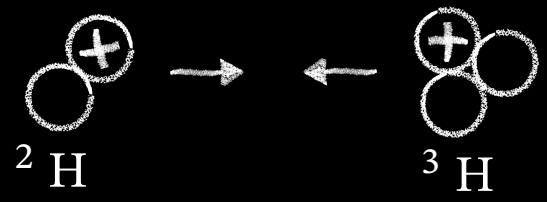
$\rho \sim 10^5 \mathrm{kg/m^3}$

$T \sim 10^8$ K

$p \sim 5 \times 10^{16}$ Pa

$\tau \sim 100$ ps

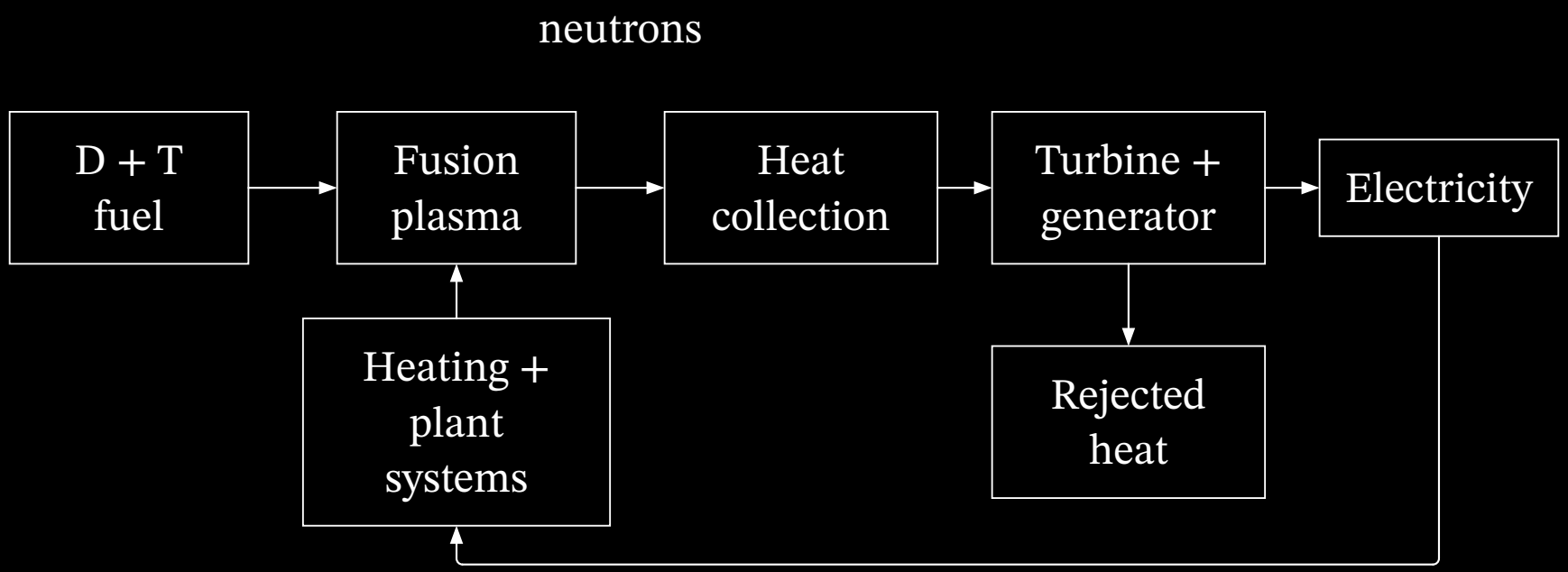
$$p\tau \sim 5 \times 10^6 \text{ Pa s}$$






NIF laser bay photograph and original annotations: FusionIntroSlides, slide 22 · LLNL/NIF · illustrative compressed-fuel parameters

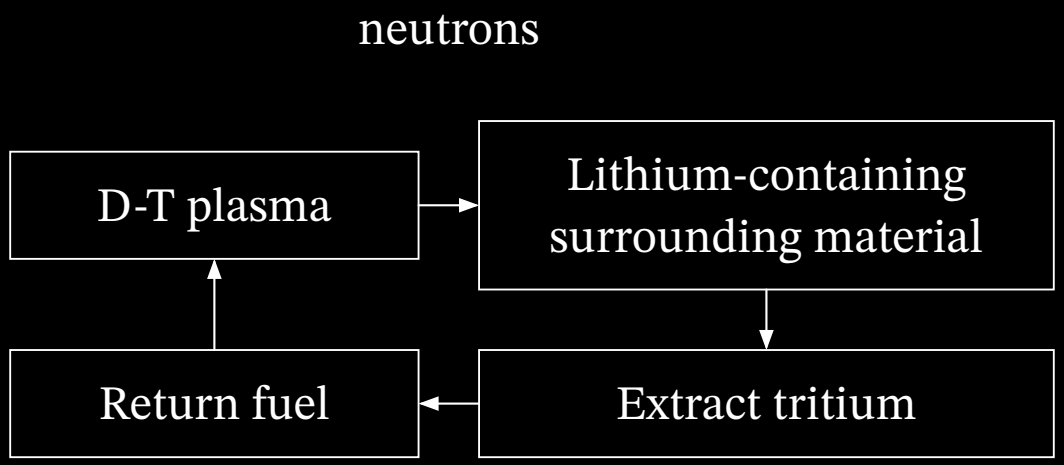
# A D-T fusion power plant




Original concept diagram; cooling, exhaust and detailed power balance developed in Chapters 4 and 10.

# Making the tritium fuel

$$n + {}^{6}\mathrm{Li} \rightarrow {}^{4}\mathrm{He} + \mathrm{T} + 4.8\ \mathrm{MeV}$$




This surrounding heat-collecting, tritium-producing region is the breeding blanket. Details: Chapter 10.

# Scientific sources


Aston · Nobel lecture (1922)

Eddington · The internal constitution of the stars (1920)

Atkinson & Houtermans · Z. Phys. 54, 656–665 (1929)

Oliphant, Harteck & Rutherford · Heavy-hydrogen reactions (1934)

Lawson · Criteria for a power producing thermonuclear reactor (1957)

NIST · Atomic weights and isotopic compositions

IAEA · Fusion energy  LLNL · NIF target chamber

# Image sources and licences

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Aston laboratory · Cambridge University Library, P42 · CC BY-NC 4.0

1932 apparatus · Cambridge University Library, P514 · CC BY-NC 4.0

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Aston plates · Isotopes (1922)

Eddington, Einstein (Arthur Sasse, 1951), Lawson, JET, NIF, Doge, Koralm: original images from FusionIntroSlides; scientific figures cited on their slides.

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AI tools assisted drafting and layout; scientific review by Christopher Albert.